

- 39** Monoclonal antibodies are produced for use in diagnosis or treatment of disease. To obtain the antibodies for an antigen, a mouse is injected with the antigen.

Some of the events in the production of the monoclonal antibody are listed.

- 1 Plasma cells are fused with cancer cells to form a hybridoma.
- 2 Hybridomas that secrete the required antibody are identified and cloned.
- 3 B-lymphocytes that recognise the antigen multiply and become plasma cells.
- 4 Hybridomas divide by mitosis and secrete antibodies.
- 5 Plasma cells are removed from the mouse spleen.

What is the sequence of the first four events in the production of the monoclonal antibody?

- A** 3 → 5 → 1 → 2
- B** 3 → 5 → 1 → 4
- C** 5 → 1 → 2 → 4
- D** 5 → 1 → 4 → 2